

Small Scale Digital Forensics

CSCI 6638/4438

Final Challenge

Tracy & Carry Devices Investigation

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Introduction

Forensic analysis performed on small-scale digital devices becomes a difficult task that takes careful consideration and critical thinking to perform effective analysis. Upon completion of forensic analysis, investigators can unfold a chain reaction of underlying plots that criminals attempt to conceal through the usage of small-scale digital devices. Group 2 was assigned with the examination and analysis of a series of small-scale digital devices seized from potential criminals suspected of theft and defacement.

Small-scale digital devices can be extracted into different file formats depending on the type of device being extracted. Group 2 was given the extracted file contents from different device types, including mobile phones, tablets, and external drives. To obtain the information on the extracted files, the team utilized a wide variety of forensic analysis tools to analyze the extracted files. The forensic analysis tools selected by Group 2 can parse the contents of the extracted files and detect any encrypted files found on the device. In conducting the investigation, Group 2 ran into issues that were properly documented along with their recommended solutions. Finally, a summary report encompassing all evidence was created to conclude the investigation.

Summary of Findings

Analysis of the devices' extracted files revealed malicious intents from both suspects whose devices were seized for the investigation. The two suspects, Carry and Tracy, used the seized devices to contact each other and organize a flash mob to serve as a distraction for both suspects to breach the security force held at the National Gallery of Art in Washington, DC. The event was organized to occur during the week of July 9th, 2012, and each suspect used the event to carry out their own nefarious purposes. Tracy used the event to steal stamps from the National Gallery, and contacted her brother, Pat, to assist with the heist in any way that he could. Carry used the event to get her tablet into the National Gallery in order to deface foreign art in the museum belonging to Majavia. Prior to the day of the flash mob, Carry has been in contact with Alex, a Krasnovian businessman who wishes to embarrass America and damage public relations through the event.

Tools & Equipment

ITEM	RESOURCE	DATE USED	PURPOSE
VirtualBox	https://virtualbox.org/	11-15-24	The application used for created virtual environments and machines.
Kali Linux VM	https://www.kali.org/get-kali/#kali-platforms	11-15-24	Virtual Machine used as a forensic workstation.
Windows 10 Laptop	N/A	11-15-24	Forensic workstation used to conduct investigation.
Autopsy 4.21.0	https://www.autopsy.com/	11-21-24	Open-source digital forensics program used to investigate external drive contents.
Notepad	N/A	11-15-24	Built-in Microsoft tool used to open text documents.
GPS Coordinates – Latitude and Longitude Finder	https://gps-coordinates.org/	12-4-24	Used to find location where photos were taken.
FTK Imager 4.7.3.81	https://www.exterro.com/ftk-product-downloads/ftk-imager-4-7-3-81	11-21-24	File viewer used for examining mobile device disk images.

Methods & Procedures

Forensic Methodology

Group 2 was given a series of disk images belonging to the relative device artifacts at the start of the investigation. First, Group 2 calculated the hash values of each zip file/disk image provided by the lead investigator. This was done to ensure the validity of each file as well as the reproducibility of this lab procedure. To examine the disk images' contents, Group 2 utilized a variety of digital forensic tools to parse data and reveal file contents. By using various digital forensic tools, Group 2 can decompile the unintelligible content of the original files into human-readable data artifacts to determine the malicious intents of the suspects.

FTK Imager is a free file preview and imaging tool that allows users to examine the contents of zip files to determine which files to select for data parsing. Once FTK Imager is installed, Group 2 selected the zip files provided to view the contents inside of the files. A file hierarchy system is provided to the user to easily navigate the file system. FTK Imager was used to locate the files to be used for data parsing, indicated by the '.bin' file extension.

Autopsy is an open-source digital forensics tool used to parse disk image files and reveal the file contents found on the disk image. Upon installation of Autopsy, users are prompted to create a case in which evidence will be placed into and analyzed. Group 2 selected all the appropriate disk images to be analyzed and separated them based on both suspect and device type. Autopsy then analyzes the contents of each disk image, highlighting key points found in the data such as email addresses and contact information.

Timeline Logs

#	Date	Time	Description
1	11-15-24	11:14 AM	Forensic analysis begun on disk images
2	11-21-24	6:30 PM	Tools installed on forensic workstation
3	11-21-24	6:37 PM	Hash values were calculated for disk images
4	11-23-24	5:50 PM	Autopsy analysis completed for disk images
5	12-4-24	11:22 PM	Forensic analysis on disk images completed

Procedures

Upon receiving the forensic images for the devices of suspects Tracy and Carry, Group 2 first calculated the hash values for the given files. Figure 1 shows the resulting hashes of each file.

```
PS C:\Users\dgcia\OneDrive - University of New Haven\Desktop\images> Get-FileHash -Algorithm SHA256 "C:\Users\dgcia\OneDrive - University of New Haven\Desktop\images\carry-phone-2012-07-15-final.zip"

Algorithm Hash Path
-----
SHA256 5CFEC4E899E7052907286934C6F98F97492985E5A48DAEB64549F96719792D9E C:\Users\dgcia\OneDrive - Uni...

PS C:\Users\dgcia\OneDrive - University of New Haven\Desktop\images> Get-FileHash -Algorithm SHA256 "C:\Users\dgcia\OneDrive - University of New Haven\Desktop\images\carry-phone-logical-2012-07-15-0618.zip"

Algorithm Hash Path
-----
SHA256 CBCE1C354884E8FA302A05A6E41C9980FC3BA252B2F74E73252162540F7357 C:\Users\dgcia\OneDrive - Uni...

PS C:\Users\dgcia\OneDrive - University of New Haven\Desktop\images> Get-FileHash -Algorithm SHA256 "C:\Users\dgcia\OneDrive - University of New Haven\Desktop\images\carry-tablet-2012-07-16-final.E01"

Algorithm Hash Path
-----
SHA256 26A6EA3049C86AFDD34862C453FC272A5AB4C64954AE51D23CF9DF688473A44B C:\Users\dgcia\OneDrive - Uni...

PS C:\Users\dgcia\OneDrive - University of New Haven\Desktop\images> Get-FileHash -Algorithm SHA256 "C:\Users\dgcia\OneDrive - University of New Haven\Desktop\images\carry-tablet-2012-07-16-final.tar"

Algorithm Hash Path
-----
SHA256 C70762E49D08F95CFD11246A3E4D1FC48A28D7182D1525B462638A28331793F C:\Users\dgcia\OneDrive - Uni...

PS C:\Users\dgcia\OneDrive - University of New Haven\Desktop\images> Get-FileHash -Algorithm SHA256 "C:\Users\dgcia\OneDrive - University of New Haven\Desktop\images\tracy-external-2012-07-16-final.E01"

Algorithm Hash Path
-----
SHA256 8FF94102154858E97057ED7064C576319CAFAC48FEAD179E070AF77C586078 C:\Users\dgcia\OneDrive - Uni...

PS C:\Users\dgcia\OneDrive - University of New Haven\Desktop\images> Get-FileHash -Algorithm SHA256 "C:\Users\dgcia\OneDrive - University of New Haven\Desktop\images\tracy-phone-2012-07-15-final.E01"

Algorithm Hash Path
-----
SHA256 71AED95A86A753DEC4EF4033ED7F52D6577C8534C8A01E83FD27683E621607 C:\Users\dgcia\OneDrive - Uni...

PS C:\Users\dgcia\OneDrive - University of New Haven\Desktop\images> Get-FileHash -Algorithm SHA256 "C:\Users\dgcia\OneDrive - University of New Haven\Desktop\images\Tracy-phone-Logical-2012-07-15-1317.zip"

Algorithm Hash Path
-----
SHA256 1E42870FF75DD2F884FF468E3EF5F31528894B64030831B442776E522D30329 C:\Users\dgcia\OneDrive - Uni...

PS C:\Users\dgcia\OneDrive - University of New Haven\Desktop\images> |
```

Figure 1: File hashes for forensic images

The given forensic images contains a variety of file extensions, including “zip”, “tar”, and “E01” files. To determine which files can be analyzed, Group 2 used FTK Imager to view the contents concealed inside the “zip” and “tar” files. “E01” files were excluded from this step of the procedure because they are direct disk image files that work with most forensic analysis tools, including the tool used in this investigation: Autopsy. When FTK Imager is launched on the forensic workstation, users are prompted to add evidence items. Group 2 selected the “zip” and “tar” files under the form of a disk image to view the file contents and find an appropriate disk image. FTK Imager adds the files to a dashboard, allowing the user to navigate the file systems at their own pace. Figure 2 shows the FTK Imager GUI with an example of an image file found in one of the zip files.

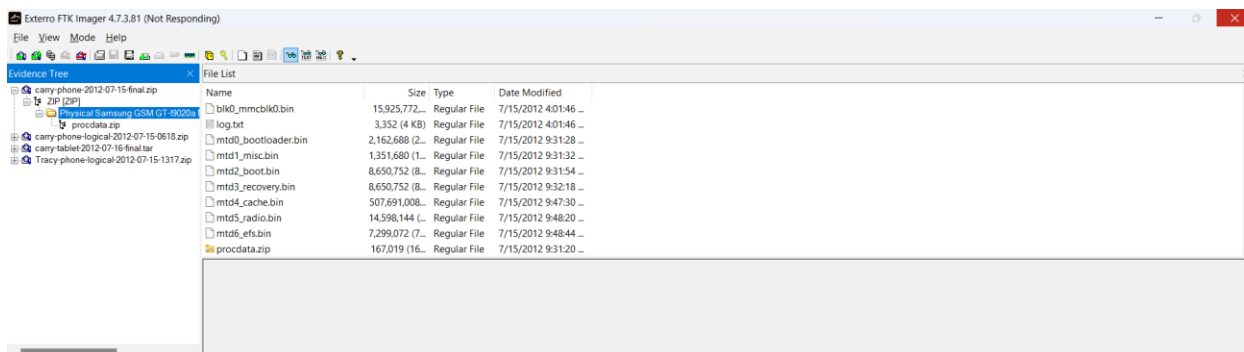


Figure 2: FTK Imager GUI with sample disk image

In Figure 2, the disk image for “carry-phone-2012-07-15-final.zip” contains a series of files with the “bin” file extension. This file extension can be usable inside a majority of forensic analysis tools. Carry’s phone is the only device that doesn’t have an E01 file, so with this information Group 2 decided to start up Autopsy to perform analysis on the provided disk images.

Autopsy asks the user to create cases in order to add pieces of evidence and begin analysis. Group 2 created a case as well as a subgroup for each device obtained for the investigation and permitted Autopsy to use all of its methods to analyze the disk images and find any suspicious details. Once Autopsy finished analysis, Group 2 was able to view the individual files of each disk image as well as the data findings provided by Autopsy. Figure 3 shows the resulting GUI provided by Autopsy once the analysis has been completed for all four devices.

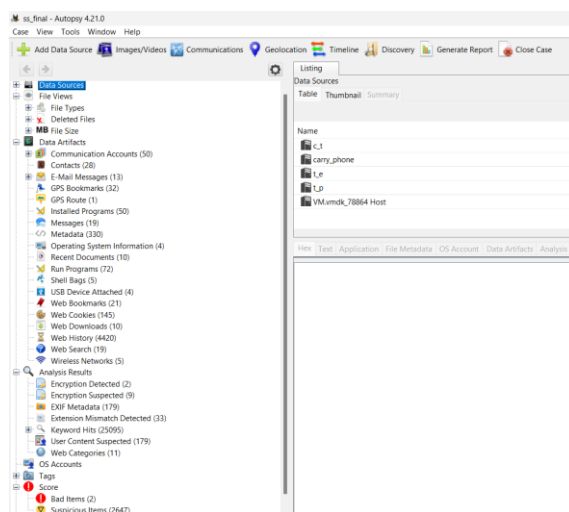


Figure 3: Autopsy GUI

Findings

Carry's Phone

Investigators were able to recover device information contained within a bug report from Carry's phone. Information from the discovered evidence included memory, processes running from the CPU, and zone information. The information was found in "Carry-Phone-Logical-2012-07-15-0618\bugreports". The first bug report was taken on 7/8/2012 at 11:00:56.

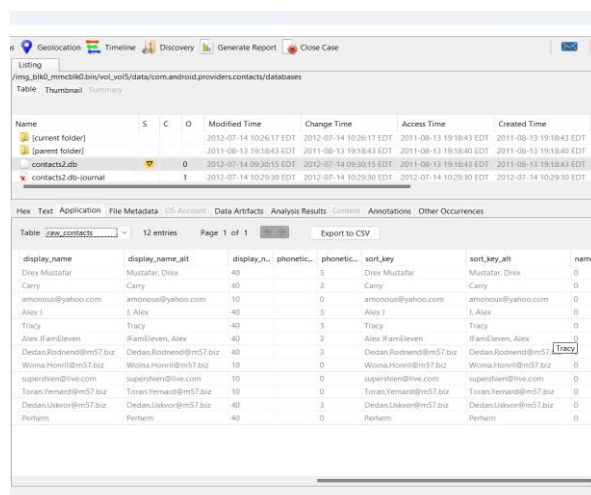
```
=====
== dumpstate: 2012-07-08 11:00:56
=====

Build: GTJ61
Bootloader: I9020XXKA3
Radio: I9020UCKE1
Network: T-Mobile
Kernel: Linux version 2.6.35.7-ge382d80 (android-build@apa28.mtv.corp.google.com) (gcc version 4.4.3 (GCC) ) #1 PREEMPT Thu Mar 31 21:11:55 PDT 2011
Command line: console=ttyFIQ0 no_console_suspend androidboot.serialno=3294C6EEFA700EC androidboot.bootloader=I9020XXKA3 androidboot.baseband=I9020UCKE1 and
roidboot.carrier=ATT gain_code=2 s3cfs.bootloaderfb=0x34a00000 mach-herring.lcd_type=0x00000000 oem_state=locked

----- MEMORY INFO (/proc/meminfo) -----
MemTotal: 353400 kB
MemFree: 102348 kB
Buffers: 11848 kB
Cached: 64904 kB
SwapCached: 0 kB
Active: 128612 kB
Inactive: 60096 kB
Active(anon): 106084 kB
Inactive(anon): 6508 kB
Active(file): 22528 kB
Inactive(file): 53588 kB
```

Figure 4: Device Information for Carry's Phone

Investigators were also able to obtain contact information for accounts that Carry has contacted using this device. Details range from names to email addresses to phone numbers. Figure 5 contains an example of the contacts data found on the device



The screenshot shows a forensic tool interface with a 'Listing' section at the top, displaying a table of files. Below this is a 'Table' section showing a list of contacts. The bottom section is a detailed table of contact entries with columns for display_name, display_name_alt, display_n, phonetic, sort_key, sort_key_alt, and name_s.

Name	S	C	O	Modified Time	Change Time	Access Time	Created Time	S
[current folder]				2012-07-14 10:26:17 EDT	2012-07-14 10:26:17 EDT	2011-08-13 19:18:43 EDT	2011-08-13 19:18:43 EDT	4
[parent folder]				2011-08-13 19:18:43 EDT	2011-08-13 19:18:43 EDT	2011-08-13 19:18:43 EDT	2011-08-13 19:18:43 EDT	4
contacts2.db			0	2012-07-14 09:30:15 EDT	2012-07-14 09:30:15 EDT	2011-08-13 19:18:43 EDT	2011-08-13 19:18:43 EDT	5
contacts2.db-journal			1	2012-07-14 10:29:30 EDT	2012-07-14 10:29:30 EDT	2012-07-14 10:29:30 EDT	2012-07-14 10:29:30 EDT	1

Table	display_name	display_name_alt	display_n	phonetic	phonetic	sort_key	sort_key_alt	name_s
display_name	Mustafar, Drex	Mustafar, Drex	40	5	Drex Mustafar	Mustafar, Drex	0	
display_name	Carry	Carry	40	3	Carry	Carry	0	
display_name	amonusu@yahoo.com	amonusu@yahoo.com	10	0	amonusu@yahoo.com	amonusu@yahoo.com	0	
display_name	Alex I.	Alex I.	40	3	Alex I.	Alex I.	0	
display_name	Tracy	Tracy	40	3	Tracy	Tracy	0	
display_name	Alex I.famEleven	Alex I.famEleven	40	3	Alex I.famEleven	Alex I.famEleven	0	
display_name	Dedan.Rodrend@m57.biz	Dedan.Rodrend@m57.biz	40	3	Dedan.Rodrend@m57.biz	Dedan.Rodrend@m57.biz	0	
display_name	Woma.Honri@m57.biz	Woma.Honri@m57.biz	10	0	Woma.Honri@m57.biz	Woma.Honri@m57.biz	0	
display_name	superthand@me.com	superthand@me.com	10	0	superthand@me.com	superthand@me.com	0	
display_name	Toran.Vernard@m57.biz	Toran.Vernard@m57.biz	10	0	Toran.Vernard@m57.biz	Toran.Vernard@m57.biz	0	
display_name	Dedan.Ukvor@m57.biz	Dedan.Ukvor@m57.biz	40	3	Dedan.Ukvor@m57.biz	Dedan.Ukvor@m57.biz	0	
display_name	Perhem	Perhem	40	0	Perhem	Perhem	0	

Figure 5: Contact information on Carry's Phone

Additionally, Carry's phone also yielded photos taken that were stored in Carry-Phone-Logical-2012-07-15-0618\DCIM\Camera. After further analysis, information from the Roaches Run Waterfowl Sanctuary in Arlington, VA 22202.

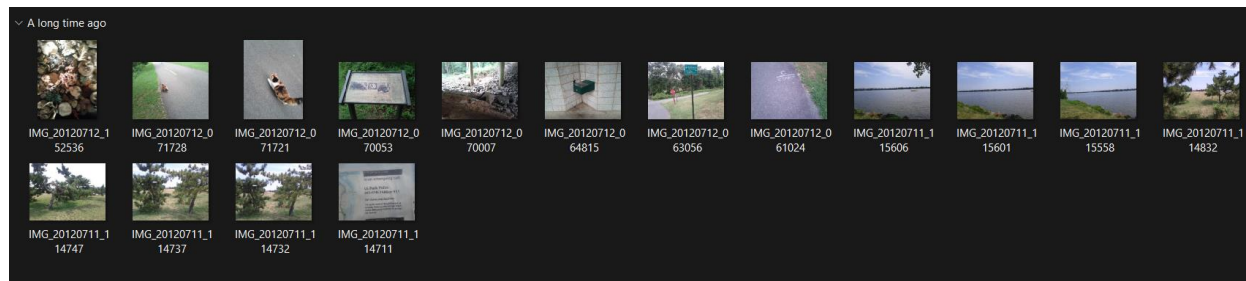


Figure 6: Photos taken from Carry's Phone

Type	Value	Source(s)
List of Waypoints	points 1 TSK_GEO_LATITUDE = 38.883797 TSK_GEO_LONGITUDE = -77.103551 2 TSK_GEO_LATITUDE = 38.864506 TSK_GEO_LONGITUDE = -77.039995 TSK_NAME = Roaches Run Waterfowl Sanctuary Arlington, VA 22202	Android Analyzer
Date/Time	2012-07-12 11:24:19 EDT	Android Analyzer
Program Name	Google Maps History	Android Analyzer
Source File Path	/img_carry-tablet-2012-07-16-final.E01/vol_vo17/data/com.google.android.apps.maps/databases/da_destination_history	
Artifact ID	-9223372036854775781	

Figure 7: Geographic Information from photos taken

Investigators were also able to find location data where Wi-Fi was used on the device in the form of latitude and longitude values. Group 2 used the tool GPS Coordinates to find the location where the latitude and longitude values are equal. The values led to a location near N Glebe Rd in Arlington, VA 22203.

Name	S	C	O	Modified Time	Change Time	Access Time	Created Time	Size	Flags(Dir)	Flags(Meta)	Kiv
[parent folder]				2011-08-13 19:18:48 EDT	2011-08-13 19:18:48 EDT	2011-08-13 19:18:33 EDT	2011-08-13 19:18:33 EDT	4096	Allocated	Allocated	unk
cache.cell	0			2012-07-14 09:26:03 EDT	2012-07-14 09:26:03 EDT	2012-07-11 08:41:23 EDT	2012-07-11 08:41:23 EDT	586	Allocated	Allocated	unk
cache.wifi	0			2012-07-14 09:26:03 EDT	2012-07-14 09:26:03 EDT	2012-07-11 08:41:23 EDT	2012-07-11 08:41:23 EDT	1024	Allocated	Allocated	unk
DATA_Preferences	1			2012-07-11 11:44:22 EDT	2012-07-11 11:44:22 EDT	2011-08-13 19:18:48 EDT	2011-08-13 19:18:48 EDT	2000	Allocated	Allocated	unk
glisplatform.aey	1			2012-07-11 08:41:23 EDT	2012-07-11 08:41:23 EDT	2012-07-11 08:41:23 EDT	2012-07-11 08:41:23 EDT	37	Allocated	Allocated	unk

Type	Value	Source(s)
Latitude	38.8815326	Android Analyzer
Longitude	-77.155995	Android Analyzer
Date/Time	2012-07-14 09:26:03 EDT	Android Analyzer
Program Name	cache.wifi Location History	Android Analyzer
Source File Path	/img_bk0_mmcblk0.bv/vol_vo15/data/com.google.android.location/files/cache.wifi	
Artifact ID	-9223372036854745734	

Figure 8: Wi-Fi usage from Carry's phone

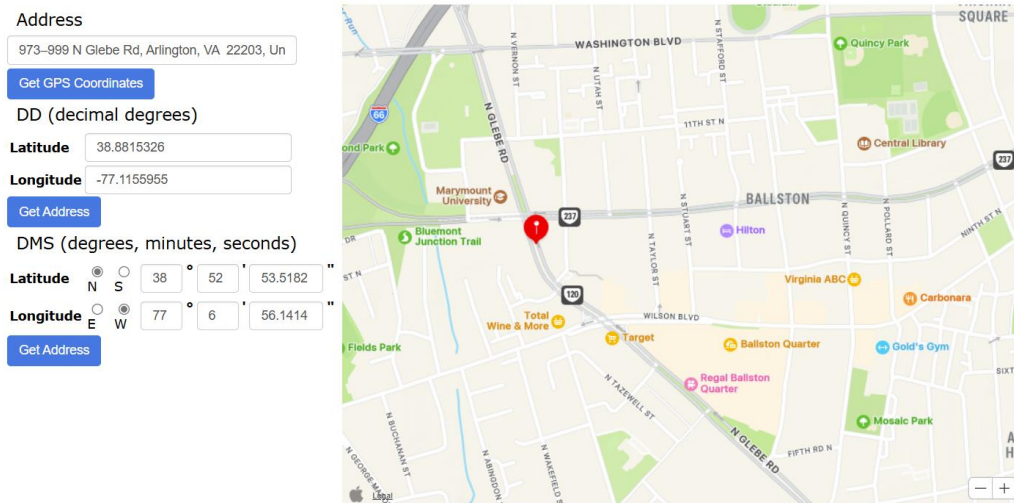


Figure 9: Location of coordinates

Carry's Tablet

Category	Details
Device Model	Asus EPAD
Hardware ID	0380624540801557
Manufacturer	Asus
Android Version	4.0.3
Build ID	IML74K
Build Date	May 3, 17:32:01 CST, 2012
Wi-Fi Interface	wlan0
Bluetooth Version	V8.30
SIM Required	No
CPU Architecture	ARMv7 (32-bit)
Storage Details	16 GB

Table 1: Carry's Tablet Device Information

Utilizing Autopsy, investigators were able to find several contacts stored on Carry's tablet. Within this discovery, names and emails were found to belong on the device. Additionally, two contacts included phone numbers. It is assumed that the contact Carry belongs to the owner of the tablet as well as Carry's Phone. This was discovered from Autopsy in the contacts analysis.

Keyword search 5 Results

Table Thumbnail Summary

Save Table as CSV

Name	Keyword Preview	Location
Unalloc_10275_571064320_1115684864	(0[h]patch/system/<build.prop,ps<ts317f"]1ul]hql	/img_carry-tablet-2012-07-16-final.E01/vol_vol2/\$U...
build.prop	<build.prop>	/img_carry-tablet-2012-07-16-final.E01/vol_vol1/buil...
build.prop-slack	<build.prop>-slac	/img_carry-tablet-2012-07-16-final.E01/vol_vol1/buil...

Hex Text Application File Metadata OS Account Data Artifacts Analysis Results Context Annotations Other Occurrences

Strings Extracted Text Translation

Page: 1 of 1 Page Matches on page: 1 of 1 Match 100% Reset Text Source: Search Results

```
[build.prop, # begin build properties
# autogenerated by buildinfo.sh
ro.build.id=IML74K
ro.build.display.id=IML74K.US_epad-9.2.1.24-20120503
ro.build.version.incremental=US_epad-9.2.1.24-20120503
ro.build.version.sdk=15
ro.build.version.codename=REL
ro.build.version.release=4.0.3
ro.build.date=Thu May 3 17:32:01 CST 2012
ro.build.date.utc=1336037521
ro.build.type=user
ro.build.user=
ro.build.host=Mercury
ro.build.tags=release-keys
ro.product.model=EPAD
ro.product.brand=asus
```

Figure 10: "build.prop" in Carry's tablet

/img_carry-tablet-2012-07-16-final.E01/vol_vol7/data/com.android.providers.settings/databases 5 Result

Table Thumbnail Summary

Save Table as CSV

Name	S	C	O	Modified Time	Change Time	Access Time	Created Time	Size
[current folder]				2012-07-15 04:44:12 EDT	2012-07-15 04:44:12 EDT	2012-07-06 10:49:59 EDT	2012-07-06 10:49:59 EDT	4096
[parent folder]				2012-07-06 10:49:59 EDT	2012-07-06 10:49:59 EDT	2012-07-06 10:49:51 EDT	2012-07-06 10:49:51 EDT	4096
settings.db				2012-07-15 04:51:52 EDT	2012-07-15 04:51:52 EDT	2012-07-06 10:49:59 EDT	2012-07-06 10:49:59 EDT	33792
settings.db-shm				2012-07-15 04:51:52 EDT	2012-07-15 04:51:52 EDT	2012-07-15 04:44:12 EDT	2012-07-15 04:44:12 EDT	32768
settings.db-wal				2012-07-15 04:51:52 EDT	2012-07-15 04:51:52 EDT	2012-07-15 04:44:12 EDT	2012-07-15 04:44:12 EDT	5272

Hex Text Application File Metadata OS Account Data Artifacts Analysis Results Context Annotations Other Occurrences

Table secure 48 entries Page 1 of 1 Export to CSV

_id	name	value
2	hardware_id	0380624540801557
3	data_roaming	0
4	install_non_market_apps	0
6	assisted_gps_enabled	1
7	network_preference	9
8	usb_mass_storage_enabled	1
10	wifi_networks_available_n...	1
11	preferred_network_mode	0
12	cdma_cell_broadcast_sms	1

Figure 11 "settings.db" in Carry's tablet

Keyword search 5 Results

Table Thumbnail Summary

Save Table as CSV

Name	Keyword Preview	Location
accounts.db	«accounts.db«	/img_carry-tablet-2012-07-16-final.E01/vol_vol7/s
accounts.db-slack	«accounts.db«-slac	/img_carry-tablet-2012-07-16-final.E01/vol_vol7/s
accounts.db-journal	«accounts.db«-journa	/img_carry-tablet-2012-07-16-final.E01/vol_vol7/s
system@app@GoogleServicesFramework.apk@da	accounts_id as _id«accounts.db«accounts.username a	/img_carry-tablet-2012-07-16-final.E01/vol_vol7/d

Hex Text Application File Metadata OS Account Data Artifacts Analysis Results Context Annotations Other Occurrences

Table accounts 4 entries Page 1 of 1 Export to CSV

_id	name	type	password
1	cat2welve@gmail.com	com.google	1/Mxq54RXI-i7ILkYiZ988ffUQmBFhA7khF54ZE8kh9V4 RMopeTeTndnEbRO2xw-YhbC
2	cat2welve@gmail.com	com.dropbox.android.account	
3	carrysum2012@yahoo.com	com.android.email	ftptb!
4	carry.sums	com.skype.contacts.sync	284151

Figure 12 "accounts.db" in Carry's tablet

Contacts 28 Res

Table Thumbnail Summary

Save Table as CSV

Source Name	S	C	O	Name	Email	Data Source	Phone Number	Email (Internet)
contacts2.db				Alex J	alexjfam11@gmail.com	carry-tablet-2012-07-16-final.E01		
contacts2.db				Alex JFamEleven	alexjfam11@gmail.com	carry-tablet-2012-07-16-final.E01		
contacts2.db			0	Carry	carrysum2012@yahoo.com	carry-tablet-2012-07-16-final.E01	+1 (202) 725-2124	
contacts2.db				Dedan.Rodnend@m57.biz	Dedan.Rodnend@m57.biz	carry-tablet-2012-07-16-final.E01		
contacts2.db				Dedan.Uskvor@m57.biz	Dedan.Uskvor@m57.biz	carry-tablet-2012-07-16-final.E01		
contacts2.db				Drex Mustafar	bubbahotep2012@hotmail.com	carry-tablet-2012-07-16-final.E01		
contacts2.db				Perhem	perhem.shien@gmail.com	carry-tablet-2012-07-16-final.E01		
contacts2.db				Toran.Yernard@m57.biz	Toran.Yernard@m57.biz	carry-tablet-2012-07-16-final.E01		
contacts2.db			0	Tracy	tracysumtwelve@gmail.com	carry-tablet-2012-07-16-final.E01	+1 (703) 340-9661	
contacts2.db				Woina.Honril@m57.biz	Woina.Honril@m57.biz	carry-tablet-2012-07-16-final.E01		
contacts2.db				amonous@yahoo.com	amonous@yahoo.com	carry-tablet-2012-07-16-final.E01		
contacts2.db				supersdien@live.com	supersdien@live.com	carry-tablet-2012-07-16-final.E01		
contacts2.db				Alex J	alexjfam11@gmail.com	carry-tablet-2012-07-16-final.E01		
contacts2.db				Alex JFamEleven	alexjfam11@gmail.com	carry-tablet-2012-07-16-final.E01		
contacts2.db			0	Carry	carrysum2012@yahoo.com	carry-tablet-2012-07-16-final.E01	+1 (202) 725-2124	
contacts2.db				Dedan.Rodnend@m57.biz	Dedan.Rodnend@m57.biz	carry-tablet-2012-07-16-final.E01		
contacts2.db				Dedan.Uskvor@m57.biz	Dedan.Uskvor@m57.biz	carry-tablet-2012-07-16-final.E01		
contacts2.db				Drex Mustafar	bubbahotep2012@hotmail.com	carry-tablet-2012-07-16-final.E01		
contacts2.db				Perhem	perhem.shien@gmail.com	carry-tablet-2012-07-16-final.E01		

Figure 13: Contact Information found in Carry's Tablet

From the figure above, the phone number and email of Carry was shown. Additionally, Tracy's phone number was also found. Using Autopsy, information regarding web history and searches from Carry's tablet were also found within the analysis.

From	To	Date	Subject/Title	Key Information
Alex J Alex.jfam11@gmail.com	Carry Carsumtwotwelve Cat2welve@gmail.com	06-15-2012	Confirming the delivery	Discussion regarding receiving materials for the operation.
Alex J Alex.jfam11@gmail.com	Carry Carsumtwotwelve Cat2welve@gmail.com	06-16-2012	Join your friends on Skype	Alex shared Skype name "alex.jfam11"
Alex J Alex.jfam11@gmail.com	Carry Carsumtwotwelve Cat2welve@gmail.com	06-27-2012	Please consider this	Shared links to steganography apps:
				Steganographie: https://play.google.com/store/apps/details?id=com.shishkin.steganographie
				StegDroid: https://play.google.com/store/apps/details?id=uk.ac.cam.tfmw2.stegdroid
Alex J Alex.jfam11@gmail.com	Carry Carsumtwotwelve Cat2welve@gmail.com	07-05-2012	Video	Shared Dropbox link to "funny video"
Mike "Drex Mustafar" Bubbahotep2012@hotmail.com	Carry Carsumtwotwelve Cat2welve@gmail.com	07-09-2012	Flash mob at the National Gallery	Planning of flash mob logistics and services offered (marketing, coordination, access).
Carry Carsumtwotwelve Cat2welve@gmail.com	Mike "Drex Mustafar" Bubbahotep2012@hotmail.com	07-09-2012	RE: Flash mob at the National Gallery	Confirmation of plan: Two groups entering through east and west entrances 12pm sharp.
Carry Carsumtwotwelve Cat2welve@gmail.com	Alex J Alex.jfam11@gmail.com	07-10-2012	Digital signatures for passports	Shared digital signatures for passports and asked for the verification of associates.
Dropbox Noreply@dropbox.com	Carry Carsumtwotwelve Cat2welve@gmail.com	07-12-2012	Bonus space received!	Confirmation of a large quantity of photos/videos uploaded to dropbox. Potentially linked to steganography.
Alex J Alex.jfam11@gmail.com	Carry Carsumtwotwelve Cat2welve@gmail.com	07-12-2012	Re: Video	Fixed files attached (passport/documents)
Carry Carsumtwotwelve Cat2welve@gmail.com	Alex J Alex.jfam11@gmail.com	07-12-2012	Re: Chat with Alex J	Confirmation of plans to meet with associates the following week.

Table 2: Significant Emails

Keyword search 4 Results

Table Thumbnail Summary

Save Table as CSV

Name	Keyword Preview	Location
Unalloc_10279_6512713728_7047479296	have attached a> >«very funny video« please take a lo...	/img_carry-tablet-2012-07-16-final.E01/vol_vol7/\$U...
mailstore.cat2welve@gmail.com.db	have attached a> >«very funny video« please take a lo...	/img_carry-tablet-2012-07-16-final.E01/vol_vol7/dat...

Hex Text Application File Metadata OS Account Data Artifacts Analysis Results Context Annotations Other Occurrences

Table messages 42 entries Page 1 of 1 Export to CSV

_id	message...	conversa...	fromAdd...	toAddres...	ccAddres...	bccAddr...	replyToA...	dateSent...	dateRece...	subject	snippet	listInfo
1	1406663...	1406663...	"Carry Car...					1341499...	1341499...	gallery rec...	Here is th...	
2	1406576...	1406576...	"Alex J" <...	"" <cat2w...				1341415...	1341415...	Chat with ...	These me...	
3	1405954...	1405954...	"Alex J" <...	"Carry Car...				1340822...	1340822...	Please co...	You My d...	
4	1406517...	1405954...	"Carry Car...	"Alex J" <...				1341360...	1341360...	Re: Please...	On Wed, J...	
5	1406519...	1405954...	"Alex JFa...	"Carry Car...				1341361...	1341361...	Re: Please...	My Dear ...	
6	1406042...	1406042...	"Google+ ...	"" <cat2w...				1340906...	1340906...	Top 3 pos...	Hi Carry! ...	
7	1405943...	1405943...	"Skype" <...	"" <cat2w...			"Skype" <...	1340812...	1340812...	Your frien...	[image:] [...	
8	1404141...	1404141...	"Carry Car...	"" <Alexjf...				1339093...	1339093...	Ugly Ame...	Hello Und...	
9	1405942...	1404141...	"Alex J" <...	"Carry Car...				1340811...	1340811...	Re: Ugly A...	Greetings ...	
10	1405940...	1405940...	"" <alexjf...	"" <cat2w...				1340809...	1340809...	Carry	Krasnovia!	
11	1405323...	1405323...	"Google P...	"" <cat2w...				1340220...	1340220...	Play All D...	[image:] [...	
16	1406765...	1406765...	"Carry Car...	"Alex J" <...				1341596...	1341596...	Had a pro...	I had to fo...	

Figure 14: "mailstore.db" in Carry's tablet

The team was able to find significant evidence of the heist/flashmob planning in Carry's Gmail mailstore database, as seen in the table above.

Source Name	S	C	O	Date Accessed	URL	Title	Comment
carry-tablet-2012-07-16-final.E01				2012-07-12 19:14:31 EDT	http://www.nga.gov/		Chrome History
carry-tablet-2012-07-16-final.E01				2012-07-12 19:14:31 EDT	http://wagle.net/gps/gps/Map/onlineMap2/?addressco...		Chrome History
carry-tablet-2012-07-16-final.E01				2012-07-12 19:14:31 EDT	http://wagle.net/gps/gps/main		Chrome History
carry-tablet-2012-07-16-final.E01				2012-07-12 19:14:31 EDT	http://www.thc.org/thc-epassport/		Chrome History
carry-tablet-2012-07-16-final.E01				2012-07-12 19:14:31 EDT	http://www.icao.int/Security/mrtd/Pages/default.aspx		Chrome History
carry-tablet-2012-07-16-final.E01				2012-07-12 19:14:31 EDT	http://www.snap2objects.com/2007/08/40_best_open_s...		Chrome History
carry-tablet-2012-07-16-final.E01				2012-07-12 19:14:31 EDT	http://forum.coppermine-gallery.net/index.php?topic=5		Chrome History
carry-tablet-2012-07-16-final.E01				2012-07-12 19:14:31 EDT	http://www.photome.de/download_en.html		Chrome History
carry-tablet-2012-07-16-final.E01				2012-07-12 19:14:31 EDT	http://www.starwoodhotels.com/westin/property/overv...		Chrome History
carry-tablet-2012-07-16-final.E01				2012-07-03 19:55:12 EDT	http://krasnovia.org/	DreamHost	Chrome History
carry-tablet-2012-07-16-final.E01				2012-07-11 19:05:16 EDT	http://www.google.com/	Google	Chrome History
carry-tablet-2012-07-16-final.E01				2012-07-12 19:14:35 EDT	http://www.google.com/chrome/android/tour	Chrome for Android - Product Tour	Chrome History
carry-tablet-2012-07-16-final.E01				2012-07-12 19:14:35 EDT	https://www.google.com/intl/en/chrome/browser/mob...	Chrome for Android - Product Tour	Chrome History
carry-tablet-2012-07-16-final.E01				2012-07-12 19:15:16 EDT	http://www.nanceestar.com/CatHaiku.html	Cat Haiku	Chrome History

Figure 15: Web History from Carry's Tablet

Source Name	S	C	O	Date Accessed	Program Name	Text	Comment	Data Source
carry-tablet-2012-07-16-final.E01				2012-07-11 20:31:56 EDT	dropbox	dropbox	Google Play Search	carry-tablet-2012-07-16-final.E01
carry-tablet-2012-07-16-final.E01				2012-07-11 22:33:15 EDT	skype	skype	Google Play Search	carry-tablet-2012-07-16-final.E01
carry-tablet-2012-07-16-final.E01				2012-07-12 12:49:53 EDT	sly stego	sly stego	Google Play Search	carry-tablet-2012-07-16-final.E01

Figure 16: Web Searches from Carry's Tablet

Additionally, evidence was found when searching for images on Carry's tablet. Using Autopsy, investigators were able to locate 92 photographs found on the tablet. These can be found in the path /img_carry-tablet-2012-07-16-final.E01/vol_vol7/media/DCIM/Camera.

name	modified time	change time	access time	created time	size	flags (unit)	flags (name)
[current folder]							
[parent folder]	2012-07-10 14:00:37 EDT	2012-07-10 14:00:37 EDT	2012-01-03 15:19:17 EST	2012-01-03 15:19:17 EST	4096	Allocated	Alloc
IMG_20120707_173855.jpg	2012-07-07 17:38:55 EDT	2012-07-07 17:38:55 EDT	2012-07-07 17:38:55 EDT	2012-07-07 17:38:55 EDT	1098685	Allocated	Alloc
IMG_20120707_174032.jpg	2012-07-07 17:40:32 EDT	2012-07-07 17:40:32 EDT	2012-07-07 17:40:32 EDT	2012-07-07 17:40:32 EDT	1088929	Allocated	Alloc
IMG_20120707_174758.jpg	2012-07-07 17:47:58 EDT	2012-07-07 17:47:58 EDT	2012-07-07 17:47:58 EDT	2012-07-07 17:47:58 EDT	1664049	Allocated	Alloc
IMG_20120707_174822.jpg	2012-07-07 17:48:22 EDT	2012-07-07 17:48:22 EDT	2012-07-07 17:48:22 EDT	2012-07-07 17:48:22 EDT	1784434	Allocated	Alloc
IMG_20120707_174837.jpg	2012-07-07 17:48:37 EDT	2012-07-07 17:48:37 EDT	2012-07-07 17:48:37 EDT	2012-07-07 17:48:37 EDT	2047468	Allocated	Alloc
IMG_20120707_174912.jpg	2012-07-07 17:49:12 EDT	2012-07-07 17:49:12 EDT	2012-07-07 17:49:12 EDT	2012-07-07 17:49:12 EDT	2014359	Allocated	Alloc
IMG_20120707_174926.jpg	2012-07-07 17:49:26 EDT	2012-07-07 17:49:26 EDT	2012-07-07 17:49:26 EDT	2012-07-07 17:49:26 EDT	1063949	Allocated	Alloc
IMG_20120707_181452.jpg	2012-07-07 18:14:52 EDT	2012-07-07 18:14:52 EDT	2012-07-07 18:14:52 EDT	2012-07-07 18:14:52 EDT	962779	Allocated	Alloc
IMG_20120707_181717.jpg	2012-07-07 18:17:17 EDT	2012-07-07 18:17:17 EDT	2012-07-07 18:17:17 EDT	2012-07-07 18:17:17 EDT	1041996	Allocated	Alloc
IMG_20120708_173139.jpg	2012-07-08 17:31:39 EDT	2012-07-08 17:31:39 EDT	2012-07-08 17:31:39 EDT	2012-07-08 17:31:39 EDT	1682261	Allocated	Alloc
IMG_20120708_173146.jpg	2012-07-08 17:31:46 EDT	2012-07-08 17:31:46 EDT	2012-07-08 17:31:46 EDT	2012-07-08 17:31:46 EDT	1731396	Allocated	Alloc
IMG_20120708_173152.jpg	2012-07-08 17:31:52 EDT	2012-07-08 17:31:52 EDT	2012-07-08 17:31:52 EDT	2012-07-08 17:31:52 EDT	1467300	Allocated	Alloc
IMG_20120708_173159.jpg	2012-07-08 17:31:59 EDT	2012-07-08 17:31:59 EDT	2012-07-08 17:31:59 EDT	2012-07-08 17:31:59 EDT	1505242	Allocated	Alloc
IMG_20120708_173212.jpg	2012-07-08 17:32:12 EDT	2012-07-08 17:32:12 EDT	2012-07-08 17:32:12 EDT	2012-07-08 17:32:12 EDT	1918361	Allocated	Alloc
IMG_20120708_173218.jpg	2012-07-08 17:32:18 EDT	2012-07-08 17:32:18 EDT	2012-07-08 17:32:18 EDT	2012-07-08 17:32:18 EDT	1714298	Allocated	Alloc
IMG_20120708_173222.jpg	2012-07-08 17:32:22 EDT	2012-07-08 17:32:22 EDT	2012-07-08 17:32:22 EDT	2012-07-08 17:32:22 EDT	1608962	Allocated	Alloc

Figure 17: Images located from Carry's Tablet camera role.

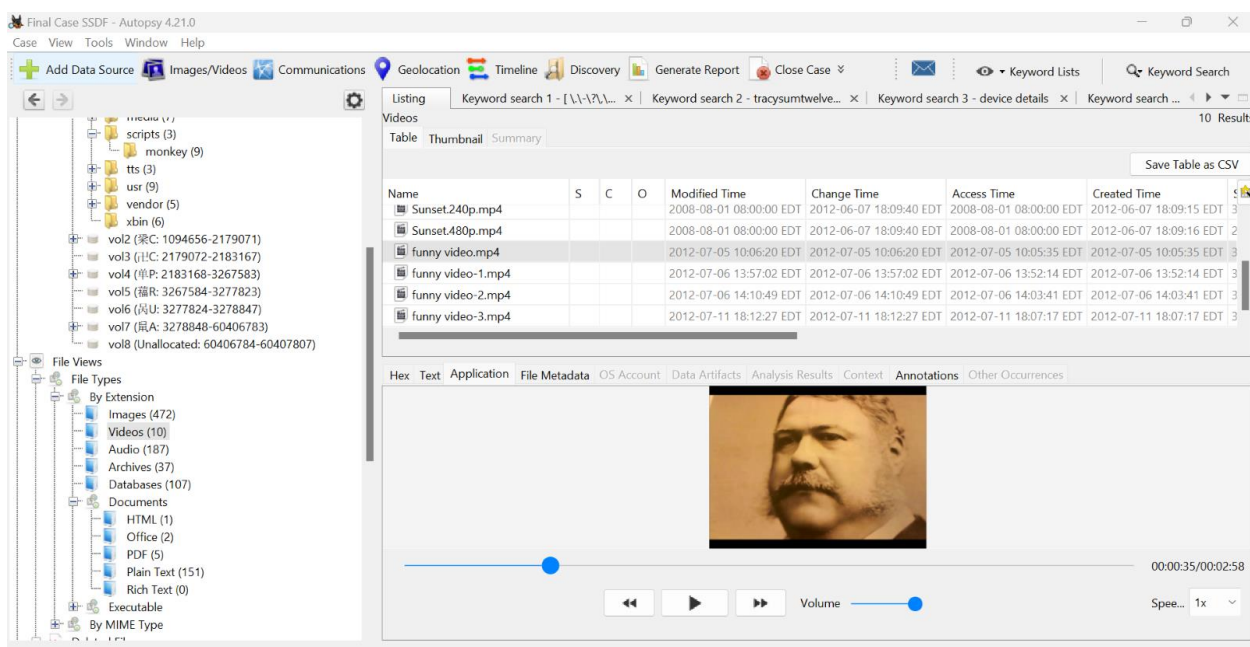


Figure 18: "funny video.mp4"

Figure 18 contains "funny video.mp4", which Alex sent to Carry through a dropbox link in an email. The video begins as an innocent skit between two talking hands until it changes at 0:35 to message from Alex, a clear attempt at hiding his message. Alex's message states:

"Hello carry, I am Alex and I am looking for your help both as a family friend and as a fellow Krasnovian. We must embarrass the United States and add tension to their relations with Majavia by defacing the rare Majavian artwork that is being put on display at the national gallery. The opportunity to strike against both America and Majavia cannot be passed up. WE may not get another chance. I understand you are a capable

individual, so I am leaving the planning to you. I will be sending some of my associates over to the states to aid you. They will arrive early this month at your international airport. Here is a picture so you can recognize them (shows picture).” In the meantime, you may contact me at alexfam11@gmail.com. Please be discrete.”

The “funny video” proves a connection between Tracy, Carry and Alex committing defamation of artwork.



Security Personnel Duty Schedule:

Shift A1 and A2 Personnel: In designated positions as of 6:00 AM till 3:00 PM F-T except for designated relief time or specific authorization via issued communication channels.

Shift B1 and B2 Personnel: In designated positions as of 6:00 AM till 3:00 PM W-SU except for designated relief times or specific authorization via issued communication channels.

Shift C Personnel: In designated positions as of 3:00 PM till 9:00 PM M-F except for designated relief time or specific authorization via issued communication channels.

Shift D Personnel: In designated positions as of 6:00 PM till 9:00 PM W-SU except for designated relief times or specific authorization via issued communication channels.

Support shift 1 Personnel: Relieve Shift A1 from 12:00 PM to 1:00 PM, Relieve Shift A2 from 1:15 pm to 2:15 PM

Support shift 2 Personnel: Relieve Shift B1 from 12:00 PM to 1:00 PM, Relieve Shift B2 from 1:15 pm to 2:15 PM

Office of Personnel Management

Figure 19: "securedownload.pdf"

The team discovered a pdf containing the National Gallery's security personnel. The pdf was suspected to be used in the planning of the flash mob and heist.

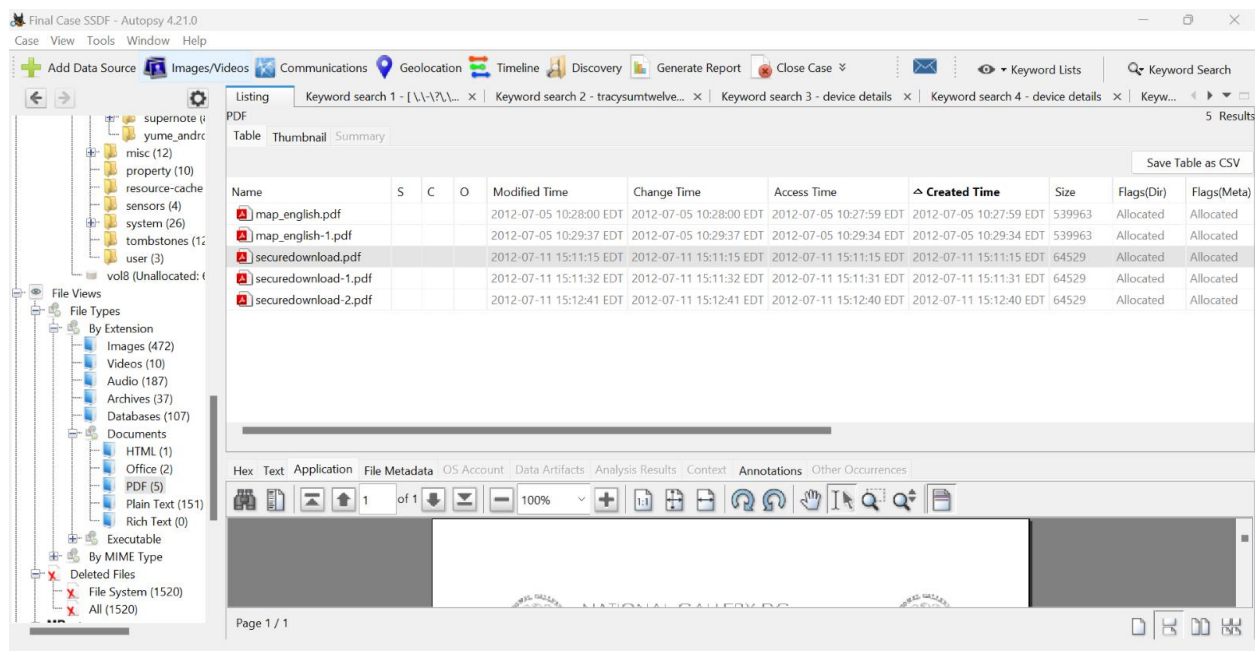


Figure 20: "map_english.pdf"

"map_english.pdf" contains a full guide and map of the National Gallery. As well, the investigators found images taken of the Gallery's security cameras.

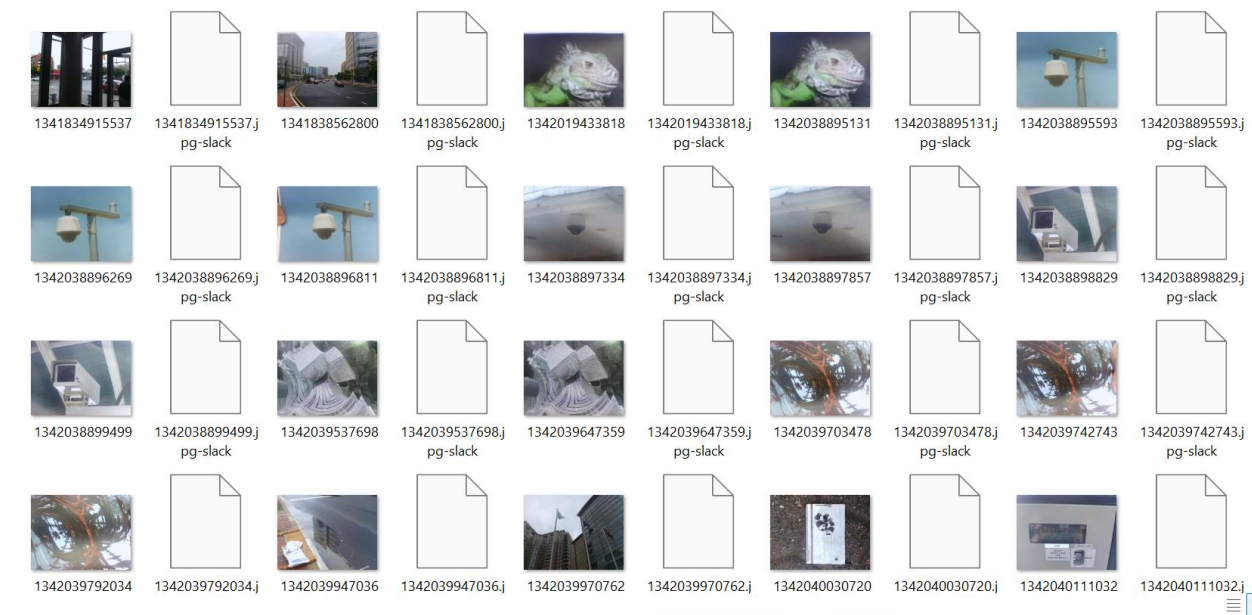


Figure 21: Security Camera Images

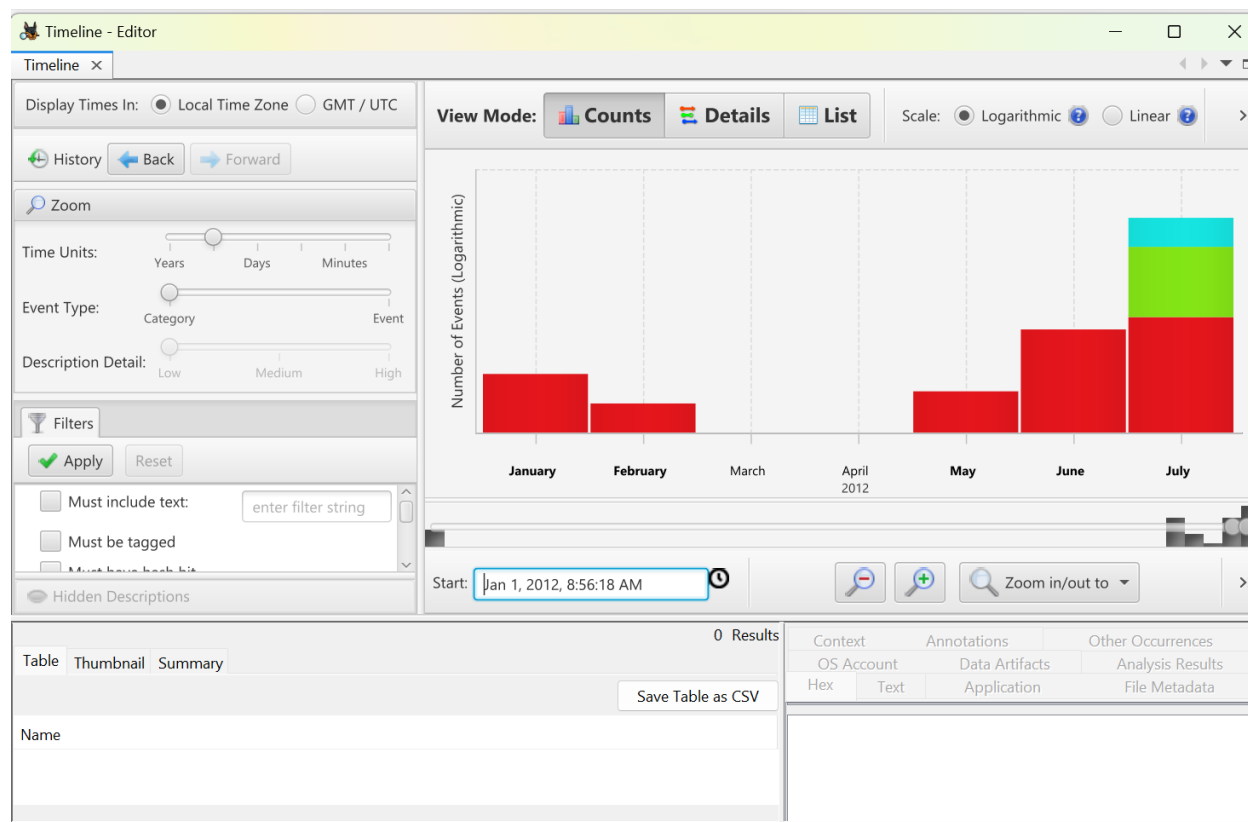


Figure 22: Activity Timeline

Figure 22 indicates that Carry's tablet has a high activity level in July, the same month where a good amount of evidence was found.

Tracy's Phone

Finding	Information	Location
Model:	iPhone 1,2	Vol5/logs/AppleSupport/general.log
OS-Version	iPhone OS 4.2.1 (8C148)	Vol5/logs/AppleSupport/general.log
Serial Number	Serial Number: 86004482Y7H	Vol5/logs/AppleSupport/general.log
Creation Time	Created: 6/6/2012 12:03:28 -0700	Vol5/logs/AppleSupport/general.log
Phone Number	1 (703) 340-9661	Vol5/logs/lockdown.log
ICCID	89014103255195342366	Vol5/logs/lockdown.log
IMEI	012021003735398	/img_tracy-phone-2012-07-15-final.E01/vol_vol5/root/Library/Lockdown/activation_records/wildcard_record.plist
Sha-256 Hash	71AED05A86A753DEC4EF4033ED7F52D6577CC B534CA0D1E83FFD27683E621607	

Table 3: Tracy's Phone information

Emails on phone (vol5/mobile/Library/mail):

- tracysumtwelve@gmail.com
- tracy.sumtwelve@nationalgallerydc.org
- coralbluetwo@hotmail.com

First, Group 2 found a series of Wi-Fi networks that the phone was connected to denoted in Figure 23.

test - Autopsy 4.21.0

Case View Tools Window Help

Directory Tree

- vol0 (Unallocated: 0-5)
- vol0 (System: 6-17408)
- SIUnalloc (1)
- JHFS+ Private Directory Data (1)
- ...JHFS+ Private Data (9)
- Applications (29)
- bin (2)
- cores (1)
- dev (1)
- Developer (1)
- Library (17)
- private (3)
- sbin (13)
- System (2)
- usr (6)
- vol0 (Data: 174086-198245)
- SIUnalloc (7)
- JHFS+ Private Directory Data (1)
- ...JHFS+ Private Data (1)
- audit (1)
- db (5)
- dhcpcdclient (3)
- launchd-db (2)
- PanicReporter (1)
- timezone (2)
- es (1)
- empty (1)
- folders (1)
- keybags (2)
- Keychain (4)
- log (3)
- logs (8)
- Managed Preferences (2)
- mobile (3)
- MobileDevice (2)
- mmap (2)
- preferences (3)
- root (2)
- run (13)
- spool (2)
- tmp (3)
- var (1)
- wireless (3)
- vol0 (Unallocated: 1982459-1982463)

Listing

Wireless Networks

Table Thumbnail Summary

Save Table as CSV

Source Name	S	C	O	SSID	MAC Address	Device ID	Device Model	Date/Time	Comment	Data Source
tracy-phone-2012-07-15-final.E01			0	MIF4620L Jetpack A801 Secure	0:15:ff:16:a9:d1			2012-06-22 14:28:40 EDT	Wifi	tracy-phone-2012-07-15-final.E01
tracy-phone-2012-07-15-final.E01			0	WhiteCheetah-guest	98:fc:11:88:62:5f			2012-07-10 18:03:47 EDT	Wifi	tracy-phone-2012-07-15-final.E01
tracy-phone-2012-07-15-final.E01			0	Silverdiner	0:24:a8:b6:7d:0			2012-07-09 15:48:52 EDT	Wifi	tracy-phone-2012-07-15-final.E01
tracy-phone-2012-07-15-final.E01			0	AMX	c3:c5:23:21:db:c3			2012-07-09 15:14:04 EDT	Wifi	tracy-phone-2012-07-15-final.E01
tracy-phone-2012-07-15-final.E01			0	NorthsideSocial	04:ff:98:ac:27			2012-07-08 16:34:59 EDT	Wifi	tracy-phone-2012-07-15-final.E01

Data Content

View Text Application File Metadata OS Account Data Artifacts Analysis Results Context Annotations Other Occurrences

Result: of Result

Figure 23: Wi-Fi networks on Tracy's phone

Group 2 found a file containing device information. The device information details what kind of operating system the device is running, along with the version of that operating system. As noted within Figure __, the group discovered that the phone is an Apple iOS device, running on iOS 6.

The screenshot shows the Autopsy 4.21.0 interface. The left sidebar displays the file tree structure, with the following items visible:

- Data Sources
 - tracy-phone-2012-07-15-final.E01_1 Host
 - tracy-phone-2012-07-15-final.E01
 - vol0 (Unallocated: 6-5)
 - vol0 (Data: 17636-1962454)
 - vol0 (Unallocated: 1962459-1962463)
- File Views
 - File Types
 - Deleted Files
 - MB File Size
 - Data Artifacts
 - Wireless Networks (3)
 - Analysis Results
 - OS Accounts
 - Tags
 - Score
 - Reports

The main window displays the 'Listing' tab for the directory `/img_tracy-phone-2012-07-15-final.E01/vol0`. The table below shows the files and directories listed:

Name	S	C	O	Modified Time	Change Time	Access Time	Created Time	Size	Flags(Dir)	Flags(Meta)	Known	MDS Hash	SHA-256 Hash
spool				2010-10-22 00:14:03 EDT	2012-06-06 14:59:48 EDT	2010-11-17 04:12:48 EST	2010-10-22 00:14:03 EDT	0	Allocated	Allocated	unknown		
tmp				2012-07-15 05:56:13 EDT	2012-07-15 05:56:13 EDT	2010-11-17 04:12:48 EST	2010-10-22 00:14:03 EDT	0	Allocated	Allocated	unknown		
vm				2010-10-12 00:28:16 EDT	2012-06-06 14:59:48 EDT	2010-11-17 04:12:48 EST	2010-10-12 00:28:16 EDT	0	Allocated	Allocated	unknown		
wireless				2012-09-12 10:55:39 EDT	2012-06-12 10:55:29 EDT	2010-11-17 04:12:48 EST	2010-10-12 00:28:17 EDT	0	Allocated	Allocated	unknown		
\$AllocationFile				0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	229376	Allocated	Allocated	unknown		
\$AttributesFile				0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	8388608	Allocated	Allocated	unknown		
\$BadBlockFile				0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0	Allocated	Allocated	unknown		
\$CatalogFile				0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	8388608	Allocated	Allocated	unknown		
\$EntentsFile				0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	4194304	Allocated	Allocated	unknown		
[parent folder]				0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0	Allocated	Allocated	unknown		

The bottom pane shows the 'Strings' tab, displaying a list of extracted text strings, including file paths and system identifiers.

Figure 24: \$CatalogFile detailing files of interests

The \$CatalogFile within Figure 24, denotes various files that were on the system. This allowed the group to peruse looking for these files and further helped to identify artifacts within the system.

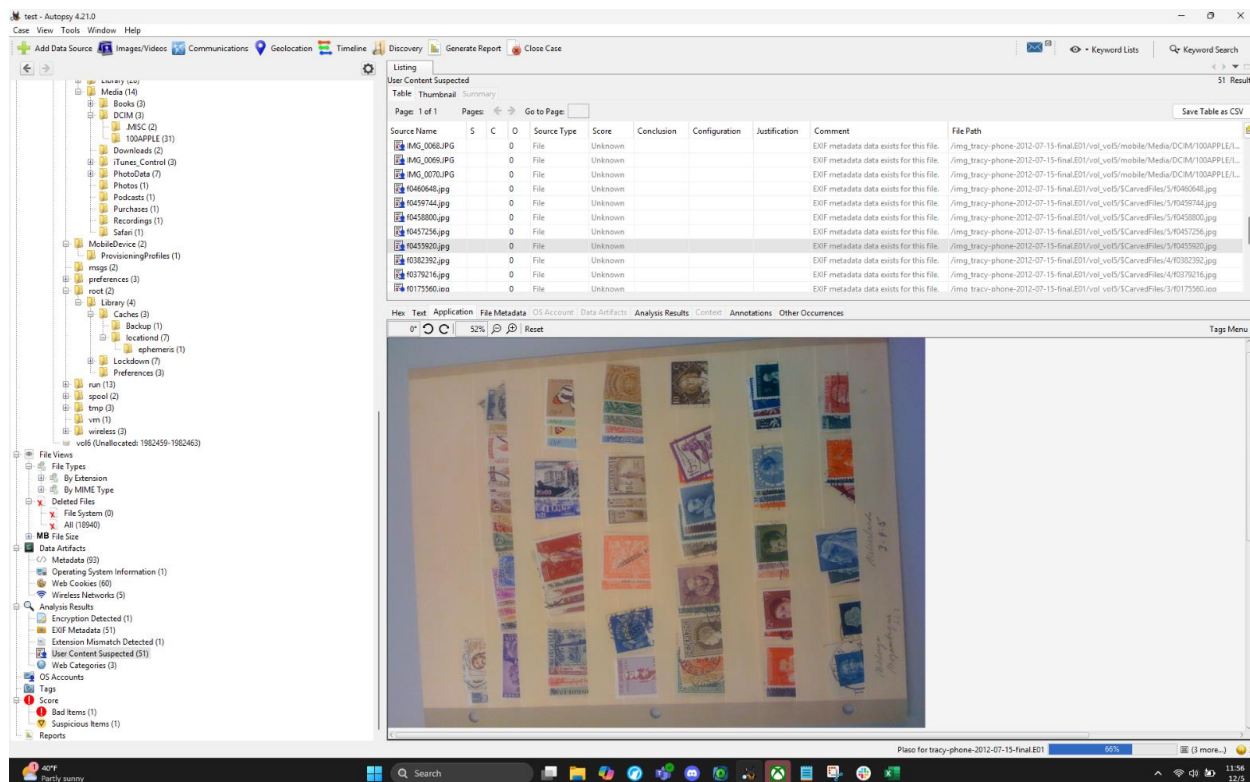


Figure 25: Stamps image on Tracy's phone

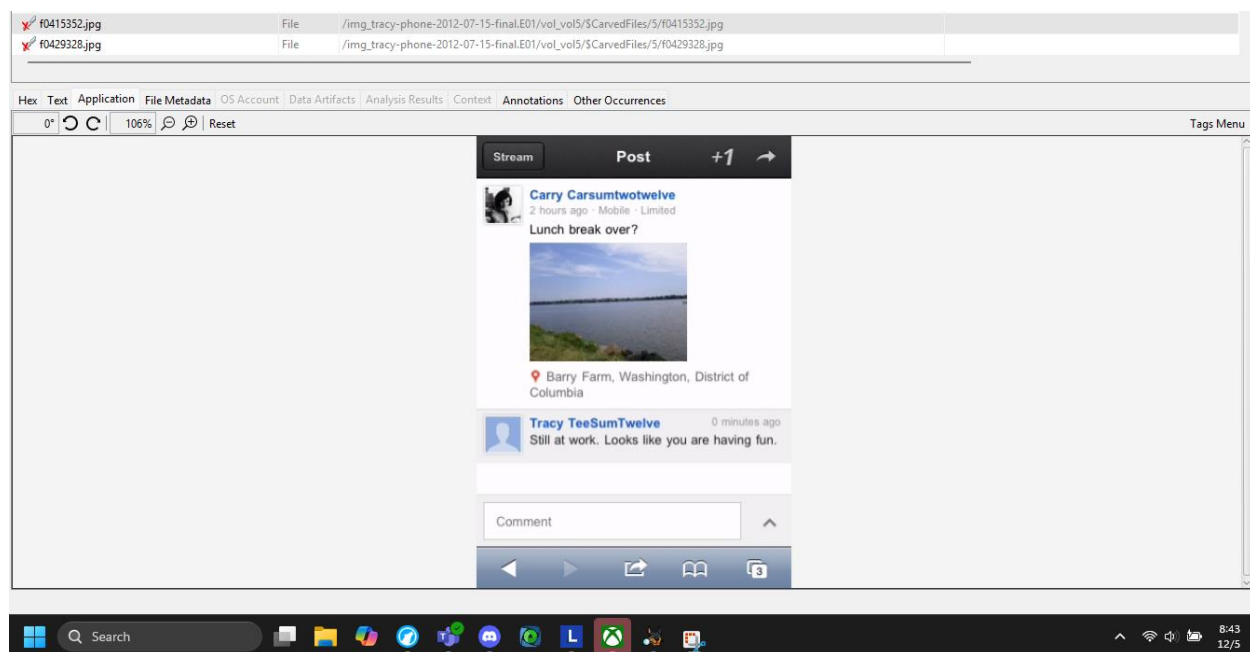


Figure 26: Carry & Tracy interaction

address	text
15713083236	What are you up to this weekend?
17038296071	I'm going out with dad after school for pizza! Thought I'd let you know if you planned to cook.
15713083236	I don't have any big plans. How about you?
17038296071	Ok, sounds good.
17038296071	Hey honey. I'm not sure if we can afford Prufrock anymore... What do you think about maybe switching to someplace else?
17038296071	moving schools at this point would be the worst! i would rather live with dad and stay at prufrock then change schools :(
12027252124	Sounds good let's shoot for one at Bubba's grill
12027252124	Okay that sounds great. See you there
15713083236	Hey can you give me a call
15713083236	Sis I'm really busy can we can do this later
15713083236	
15713083236	Ok ok I'll call in 5
12027252124	I have a table inside
12027252124	Okay brt
12069100932	Congratulations, your entry in last months drawing won you a FREE \$1,000 Target Giftcard! Enter 703" at www.target.com.trdt.biz to tell us where to ship it"
15713083236	hey sis yo friend coral got a email the attachment needs to be changed to pdf let her know
15713083236	Sure thing I'll get on it
17038296071	Going to lunch. You want to go?????
17038296071	Back at work
17038296071	I'm busy. Maybe this weekend if dad isn't busy
12027252124	I'm almost there where should I meet you?
12027252124	Just meet me out front, I'll take the tablet in.
12027252124	How's the flashmob going
17038296071	I really want to go to Dad's this weekend. He said he'll take me shopping for school

Figure 27: Conversation between Tracy and Terry

The screenshot displays the Autopsy 4.21.0 interface. On the left, a file tree shows the 'Library' folder expanded. The main pane on the right shows a table of call history. Below this, a table titled 'call' shows 5 entries with columns: ROWID, address, date, duration, flags, id, name, country. The bottom status bar shows '42°F Mostly clear' and '8:21 11/23'.

Name	S	C	O	Modified Time	Change Time	Access Time	Created Time	Size	Flags(Dir)	Flags(Meta)	Known	Location
(parent folder)				2012-06-06 15:03:43 EDT	2012-06-06 15:03:43 EDT	2010-11-17 04:13:40 EDT	2010-10-12 00:28:17 EDT	0	Allocated	Allocated	unknown	/img_tracy_phone-2012-07-15-final.ED7/vol
call_history.db				2012-07-06 11:22:54 EDT	2012-07-06 11:22:54 EDT	2012-06-06 15:03:43 EDT	2012-06-06 15:03:43 EDT	20872	Allocated	Allocated	unknown	/img_tracy_phone-2012-07-15-final.ED7/vol

ROWID	address	date	duration	flags	id	name	country...
1	6508870260	1339531480	20	4	-1		310
2	7038296191	1339534334	56	4	-1		310
3	+15713083236	1339604853	0	5	-1		310
4	5712458517	1340386486	0	4	-1		310
5	5713083236	1341507930	244	4	-1		310

Figure 28: Call details on Tracy's phone

sender	subject	to	cc	bcc
Pat.TecSumTwelve <patsumtwelve@gmail.com>	Paris Spent and answer	Tracy.TecSumTwelve <tracysumtwelve@gmail.com>		
Perry.Patsum <perry.patsum@yahoo.com>	Link me up sometime	tracysumtwelve@gmail.com <tracysumtwelve@gmail.com>		
Perry.Patsum <perry.patsum@yahoo.com>	hey sis	tracysumtwelve@gmail.com <tracysumtwelve@gmail.com>		
csqgd@mailme7.biz	Weekend plans	tracysumtwelve@gmail.com		
Tracy.SunTwelve <tracy.suntwelve@gmail.com>	Re: Regarding Terry	Tracy.SunTwelve <tracysumtwelve@gmail.com>		
TripScores < 51676@jud.stuff.bestatopre.de>	+tut-876-Your transition creditScore may have updated-sineaaap-3w	<certified@gmail.com>	<coracle.phd@gmail.com>	
The.Osion <diaguch@theosion.com>	The Osion Daily Dispatch - July 03, 2012	tracysumtwelve@gmail.com		
* Independence Day deals <P-51676@socialbugbo.socialtablee.it>	Come to direct buy for all your furniture needs this 4th of July. Get your Visitor's Pass Today!	<nationofr1_mail@gmail.com>	<nation.cop@gmail.com>	
Google+ team <rooreply.475ba20@plus.google.com>	Someone you might know on Google+	tracysumtwelve@gmail.com		
The.Osion <diaguch@theosion.com>	The Osion Daily Dispatch - July 04, 2012	tracysumtwelve@gmail.com		
Pat.TecSumTwelve <patsumtwelve@gmail.com>	Long time no see...	tracysumtwelve@gmail.com		
The.Osion <diaguch@theosion.com>	Re: Good News	Tracy.SunTwelve <tracysumtwelve@gmail.com>		
Pat.TecSumTwelve <patsumtwelve@gmail.com>	Re: Good News	tracysumtwelve@gmail.com		
confirmflow@backinaction.portall.cabtech.eduroaster.it*	You have received \$100 - \$1000 cashDeposited	Tracy.TecSumTwelve <tracysumtwelve@gmail.com>		
The.Osion <diaguch@theosion.com>	The Osion Daily Dispatch - July 09, 2012	<grnd-m01.fonotyp.scribdetex.com>		
Carry.SunTwelve <carrysun2012@yahoo.com>	Re: Long time no see...	tracysumtwelve@gmail.com		
The.Osion <diaguch@theosion.com>	Re: Long time no see...	tracysumtwelve@gmail.com		
Carry.SunTwelve <carrysun2012@yahoo.com>	The Osion Daily Dispatch - July 10, 2012	tracysumtwelve@gmail.com		
The.Osion <diaguch@theosion.com>	The Osion Daily Dispatch - July 11, 2012	tracysumtwelve@gmail.com		
Carry.SunTwelve <carrysun2012@yahoo.com>	Re: Long time no see...	tracysumtwelve@gmail.com		
Google+ <rooreply.46447a1@plus.google.com>	Carry.Carrysumtwelve added you on Google+	tracysumtwelve@gmail.com		
Carry.Carrysumtwelve <Google+?+ccgto-746d3d22@plus.google.com>	Carry.Carrysumtwelve is sharing with you on Google+	tracysumtwelve@gmail.com		
Wuina.Huon@me7.biz	Site	tracysumtwelve@gmail.com		
Renad.Lun@me7.biz	Site	tracysumtwelve@gmail.com		
Carry.Carrysumtwelve <Google+?+ccgto-746d3d22@plus.google.com>	Carry.Carrysumtwelve is sharing with you on Google+	tracysumtwelve@gmail.com		
Carry.Carrysun2012@yahoo.com	Re: Long time no see...	Tracy.TecSumTwelve <tracysumtwelve@gmail.com>		
Carry.Carrysun2012@yahoo.com	Re: Long time no see...	Tracy.TecSumTwelve <tracysumtwelve@gmail.com>		
Microsoft Office <mscscad@redy.digitriver.com>	Only 36 days left: Start a free Office training course now.	Coral.Blue.Tec <corablue2@hotmail.com>		
Pat.TecSumTwelve <patsumtwelve@gmail.com>	Fixed: can't pass up	corablue2@hotmail.com		
Tracy.SunTwelve <tracysumtwelve@gmail.com>	ding	corablue2@hotmail.com		
Pat.TecSumTwelve <patsumtwelve@gmail.com>	can't pass up	Shone.5966@hotmail.com	corablue2@hotmail.com	
Wuina.Huon@me7.biz	Busy	corablue2@hotmail.com		
Balak.Detan@me7.biz	Busy	corablue2@hotmail.com		
Tener.Aron@me7.biz	Busy	corablue2@hotmail.com		
Unshah.Torak@me7.biz	Busy	corablue2@hotmail.com		
Unshah.Torak@me7.biz	Busy	corablue2@hotmail.com		
Othman.Ang@me7.biz	Busy	corablue2@hotmail.com		
Awen.Throssam@me7.biz	Busy	corablue2@hotmail.com		
Balak.Detan@me7.biz	Busy	corablue2@hotmail.com		
Perry.Patsum <perry.patsum@yahoo.com>	Re: Some good news	Coral <corablue2@hotmail.com>		
Perry.Patsum <perry.patsum@yahoo.com>	Re: Some good news	Coral <corablue2@hotmail.com>		
Perry.Patsum <perry.patsum@yahoo.com>	Re: Some good news	Coral <corablue2@hotmail.com>		
Perry.Patsum <perry.patsum@yahoo.com>	Re: What's going on	Coral <corablue2@hotmail.com>		
Perry.Patsum <perry.patsum@yahoo.com>	Re: What's going on	Coral <corablue2@hotmail.com>		
Perry.Patsum <perry.patsum@yahoo.com>	What's going on	Coral.Blue.Tec <corablue2@hotmail.com>		
Perry.Patsum <perry.patsum@yahoo.com>	Re: Crapdone by the VHS	Coral.Blue.Tec <corablue2@hotmail.com>		
Perry.Patsum <perry.patsum@yahoo.com>	Re: ???	Coral.Blue.Tec <corablue2@hotmail.com>		
Perry.Patsum <perry.patsum@yahoo.com>	Crapdone by the VHS	Coral.Blue.Tec <corablue2@hotmail.com>		
Microsoft Office <mscscad@redy.digitriver.com>	Only 36 days left: Start a free Office training course now.	Coral.Blue.Tec <corablue2@hotmail.com>		
Coral <corablue2@hotmail.com>	Re: Busy	Awen.Throssam@me7.biz <Awen.Throssam@me7.biz>		

Figure 29: Email messages on Tracy's phone

URL	Title	Last Visited Date	Visit Count	Redirect URLs
Google+	Google+	363811154.4	2	None
Google+ Notifications	None	363811063.5	2	Google+
Google+ Login	Google+	363810953.1	1	None
Google+ Login Redirect	None	363810951.9	1	Google+ Login
Google Login	None	363810950.7	1	Google+ Login Redirect
Gorgonzola Search	gorgonzola - Google Search	363810858.3	2	None
Gorgonzola Sauce Recipe	Gorgonzola Sauce Recipe : Ina Garten : Recipes : Food Network	363795516.1	1	Food Network
Allrecipes Search	Search - Allrecipes.com	363795459.2	1	None

Biscochitos I Recipe	Biscochitos I Recipe - Allrecipes.com	363286194.6	1	Biscochitos Mobile
Mexican Cookie Recipes	Mexican Cookie Recipes - Allrecipes.com (Pg. 1)	363286163.4	1	Mexican Cookie Recipes
Mexican Cookie Search	mexican cookie - Google Search	363286154.8	1	None
Google	Google	363204567.0	2	None
Bring out yer dead!	Bring out yer dead! Maryland has no law requiring proper corpse disposal - National Strange News	363202060.5	1	None
Wacky Politics	wacky politics	363200481.1	1	None
Amazing Spider-Man Review	'The Amazing Spider-Man' really is pretty amazing - National Fanboys	363199508.0	1	None
Summer Movies	Summer Movies	363198806.4	1	None
Why Pets Are Bad Gifts	Why pets do not make good surprise presents - Chesapeake animal welfare	363198740.5	1	Why Pets Do Not Make Good Gifts
Good Surprise Presents Search	good surprise presents - Google Search	363198702.7	1	None

Table 4: Web History on Tracy's phone

Tracy's External Drive

#	From	To	Message
1	Carry(Perry)	Tracy(Coral)	Great, Now that we have everything set up it would be better to do most of our communication here and on your new "setup". This might keep us a little bit safer. I know money is rough for both of us, so we may have to 'push the envelope' a bit. A few friends from around the office are really food about these types of things. If I find out anything interesting I will shoot you an email. In the meantime let's try to shoot some ideas and back and forth.
2	Tracy(Coral)	Carry(Perry)	I know what you mean. If anything comes up around the office that we can maybe... get in on, please lets try to do so. Kiddo is getting really bent out of shape about possibly having to switch schools. I have been paying some more attention to the memos and papers that come across my desk. We get a bunch of insurance type documents that place values on a certain objects. If anything stands out, I'll let you know.
3	Tracy(Coral)	Carry(Perry)	I think I may have come across something interesting. Everybody around the office seems to be buzzed about a foreign exhibit that is supposed to be coming over. There hasn't been any official release in writing but we have been going through quite an ordeal with all this paperwork. From what...
4	Tracy(Coral)	Carry(Perry)	Okay, so there has been a lot of money being thrown around to get this exhibit over to us. I've been swimming through paperwork and memos all day, but none of it seems to have any solid information that could help us. I had to scan and verify some shipping information, and compared to what it usually costs to ship exhibits... this ones seems relatively cheap. I'm not sure if this is good or bad for us (or even an error), but I will be able to find out more later in the week.
5	Carry(Perry)	Tracy(Coral)	That may just be exactly what we need. Try to collet as much information on the exhibit as you can, the more information... the better. In the meantime, I will try to look into our options for acutally pulling something like this off.
6	Tracy(Coral)	Carry(Perry)	I was told that we are supposed to be receiving a rare colleciton of stamps. That would explain why the shipping information looked a bit out of the ordinary to me. I'm not certain of the specifics for the stamps, but they seem to be very highly valued by somebody. Maybe this is our ticket.
7	Tracy(Coral)	Carry(Perry)	Hey Perry, Here are those documents I talked to you about. The password is your old dog's name.

Table 5: Email interaction between aliases for Tracy and Carry.

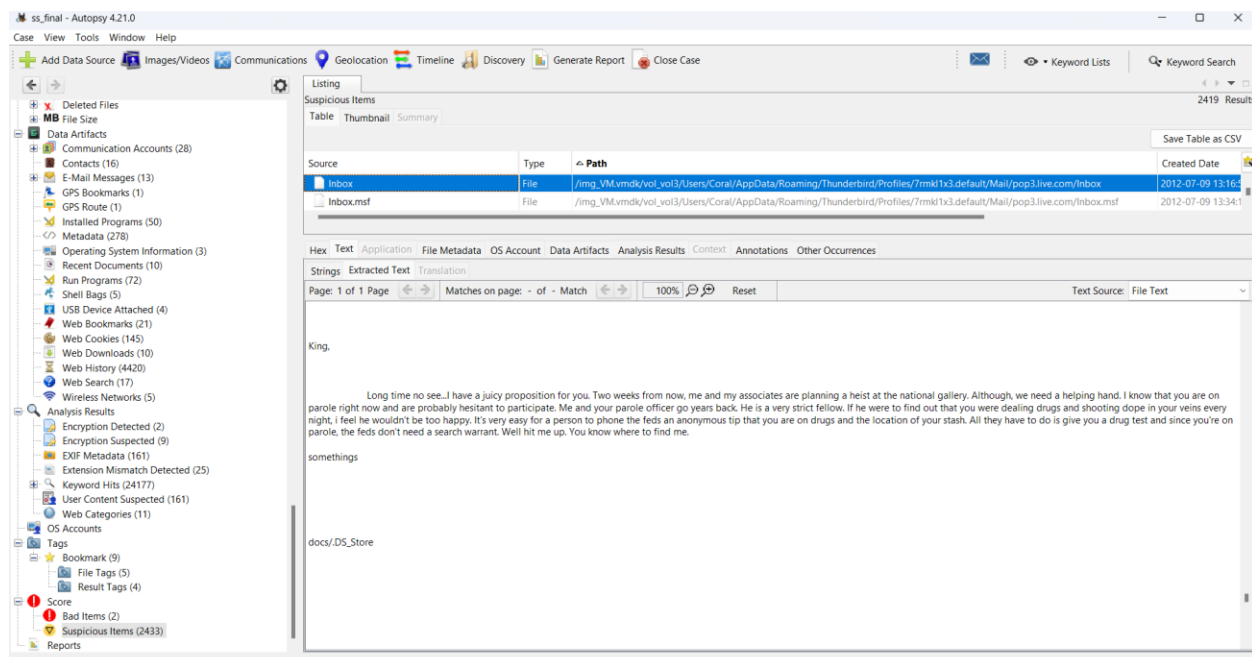


Figure 30: Email to a malicious actor named King

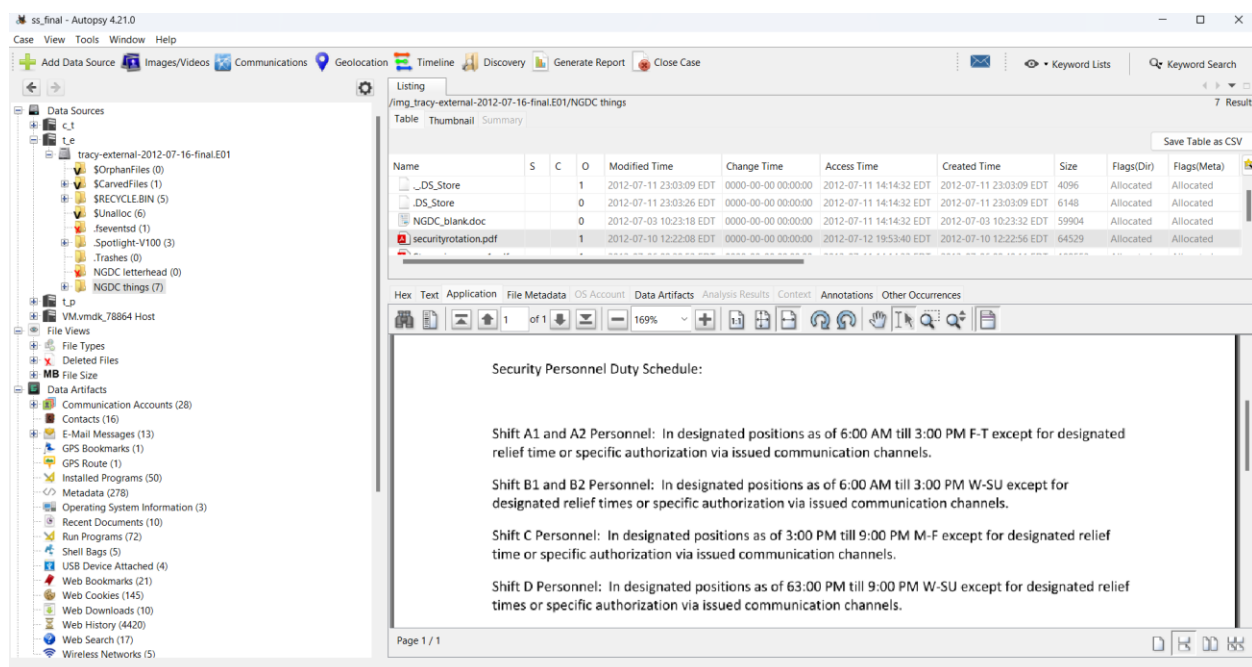


Figure 31: Security Schedule on Tracy's external drive

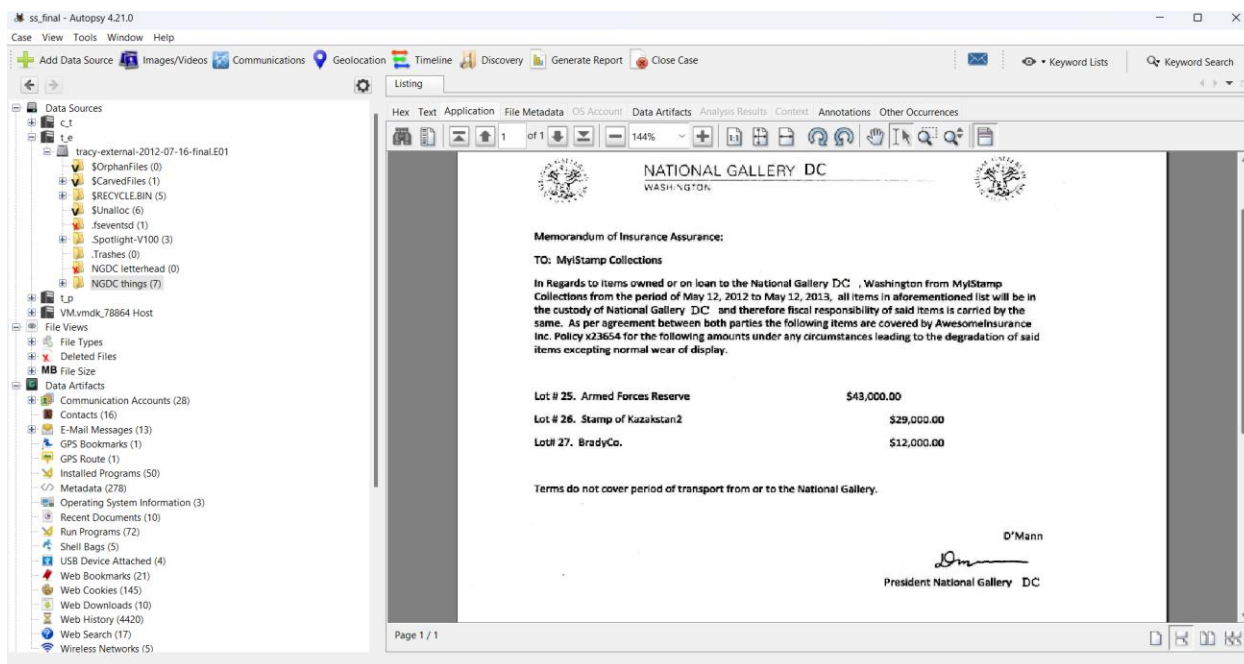


Figure 32: Stamp orders on Tracy's external drive

Challenges & Responses

Throughout the investigation, issues arose due to the functionality of some of the tools. It became difficult to properly leverage the Autopsy software to our advantage. Our method of correcting this issue came from tailoring each of our analyses to the specific evidence source that was provided.

Additionally, some of the images are not functioning properly. Initially, it was believed this was due to some sort of error in the download process, as the files were of a larger size. This was evident when dealing with Carry's phone image. Contained within the zip file was an Autopsy case file that would not open within the software. This became a challenge when examining that file and its subfolders and contents. Investigators tried to manually investigate the result of the zip. Although information was able to be recovered, it was not as effective as a functioning Autopsy analysis would've been.

Lastly, the investigators ran into issues solving the decryption of certain images. The team ran an analysis on any suspicious media files and were able to find hidden data in one (pictured below). Using Zsteg the team was able to find a file named "PGP Secret Key" hidden within the image's bit path. So, they inspected further into the file by checking the exact bit pattern "b4, b, msb, xy" and by doing so found a suspected passphrase "?pc#f88p". While the team tried many avenues and methods to use the passphrase, they were unable to find its significance.

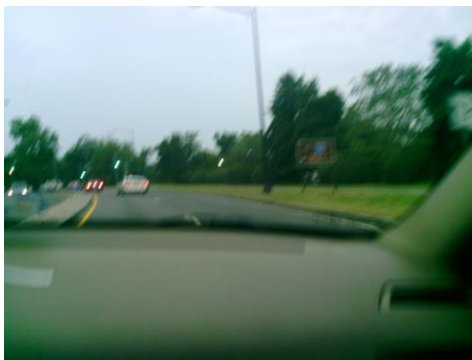


Figure 33: Image containing hidden "PGP Secret Key"

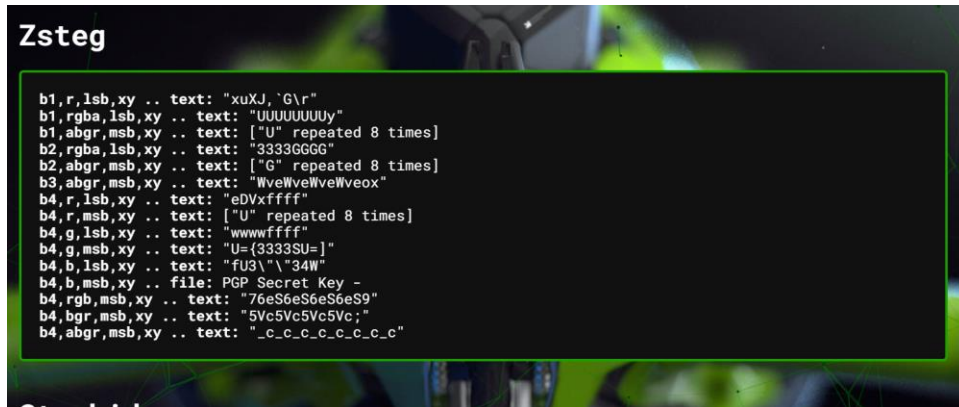


Figure 34: Zsteg analysis

Conclusion & Recommendations

Throughout the forensic investigation the team revealed substantial evidence of an organized effort to deface and steal art at the National Gallery of Art by creating a flash mob event as a diversion. The investigators uncovered many emails between Carry, Tracy and Alex discussing their detailed plans to coordinate the flash mob, heist and defacement. Alex J, a Krasnovian Businessman, initially sends Carry his plans through a hidden “funny” video, asking Carry for help. Carry’s tablet contained numerous incriminating emails, GPS data near or at the National Gallery, and searches related to their heist. Tracy’s phone held planning with Carry, evidence of a planned theft of expensive rare stamps, and the use of aliases. It was clear to the investigators that there were definite criminal activities acted out by Alex, Carry and Tracy.

The lab successfully puts an emphasis on the potential practical use of tools that were used in this lab, in the cyber forensic field. The case highlights the challenges of encrypted communications and hidden files. Through the use of their forensic expertise, the team not only uncovered the ploy against the National gallery but also gained essential experience. The group gained invaluable forensic skills which will be taken forward in their future careers and experiences as forensic investigators. careers.

References

Not Available.

Appendices

Not Available.